Seed Moth Detection from Pheromone Trap Images

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1. Problem framing

I treat the assignment as a combined **object detection and counting** task. For each trap image, the goal is to find moths and estimate how many are present. Counting is therefore not a separate model head in this proof of concept but it is simply the number of predicted boxes after non-maximum suppression, which removes duplicate overlapping detections and keeps the highest-confidence box for each object.

This framing matches the data we have. Pheromone-trap images are cluttered and noisy: glue artifacts, debris, overlapping insects, variable lighting, and perspective effects can all occur in the same frame. A box detector is a practical first step because it gives both object locations and counts without needing pixel-level labels.

2. Reference Material and Data Situation

The available reference material consists of public images of adult *Stenoma catenifer* from several online sources, including iNaturalist observations, USDA/IDtools screening aids and other [1–11]. These images are useful for species familiarisation and for understanding coarse visual priors, but they are not a curated training dataset for trap-image detection.

In practice, I collected 18 target reference images that helped establish the target appearance: elongated resting posture, brown or yellowish-tan coloration, folded wings, small apparent size, and dark wing markings. They are visually heterogeneous, often blurred, and taken against different backgrounds, so they are not representative of a single trap domain.

The project therefore separates the data into a few roles:

- **Reference target images**: public images used for species familiarisation and as the seed set for bounding-box coordinates.
- **Derived masks and cutouts**: bbox-guided masks and transparent cutouts extracted from the reference images.
- **Reviewed cutouts**: a sparse human-corrected subset used when the automatic cutout quality is poor.
- **Generated backgrounds**: generated trap-like scenes with texture, glare, stains, and debris.
- **Synthetic training images**: composed scenes with known labels for detector training.
- **Real images**: unlabeled images used only for qualitative sanity checks.

This is a deliberate bootstrapping strategy. It lets the project move forward from a small number of seed annotations, while making it clear that the current results come from a limited setup and not from a fully labeled dataset. For synthetic composition, reviewed cutouts are preferred when available, while the automatic cutouts remain as fallback inputs. This makes it possible to fix only the poor extractions rather than manually curate every single example.

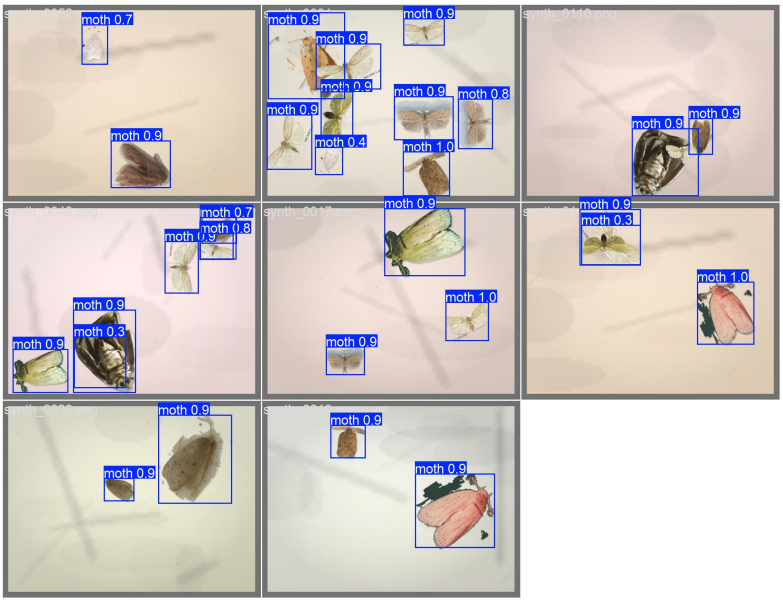
3. Method

The implemented pipeline consists of the following stages:

1. manual bounding-box coordinate annotation on the public reference images
2. bbox-guided mask and cutout extraction from those coordinates
3. sparse manual review of poor cutouts and mask corrections
4. generation of trap-like backgrounds
5. synthetic image composition with automatically generated bounding-box labels in YOLO format
6. fine-tuning of a small pretrained YOLO11n baseline
7. inference and count estimation on new images

Foreground masks are used only as an internal preprocessing step to produce cleaner cutouts for synthetic composition. The mask extraction step operates on the bbox-guided crop and is not the final task output.

(a) Synthetic training samples.



(b) Held-out synthetic validation.

(c) Real trap image with 21 detected objects.

Figure 1: Representative outputs from the PoC. The synthetic examples illustrate the compositing strategy, the validation preview shows detector output on held-out synthetic data, and the real image is a qualitative sanity check only.

Figure 1 shows two synthetic training samples, one held-out synthetic validation preview, and one real-image sanity check in a single composite panel. The synthetic generator places multiple moth instances per image, including repeated use of the same cutout, so that the detector sees denser scenes instead of a simplistic one-object-per-image regime. In the current run, the synthetic dataset contains 120 images with a 96/24 train-validation split. The synthetic scenes in this run contain between 2 and 10 moths per image, so the detector has to handle small multi-object scenes rather than a single-object setting. The output of the detector is a set of bounding boxes with confidence scores, preview images with boxes drawn on top, and a count per image. The real trap images are not used for quantitative evaluation; they are only inspected to verify that the model produces plausible detections on an unseen domain.

4. Experimental results

Table 1 summarizes the evaluation computed on held-out synthetic validation images at IoU threshold 0.5. The detector achieves high box-level agreement with a low count error per image.

Table 1: Validation metrics on held-out synthetic data only.

Images No.	GT	Pred.	TP	FP	FN	Prec.	Rec.	F1	IoU	MAE	RMSE
24	115	114	109	5	6	0.956	0.948	0.952	0.913	0.375	0.677

GT = Ground Truth boxes; Pred. = Predicted boxes; TP = True Positives; FP = False Positives; FN = False Negatives; IoU = Intersection over Union; MAE = Mean Absolute Error; RMSE = Root Mean Squared Error.

5. Discussion

The synthetic validation results are encouraging, but they should be read as an internal consistency check rather than evidence of field performance. The validation split is built from the same cutout library and background generation logic used during training, so the result mainly shows that the detector can learn the composited synthetic scenes. The remaining gap is the transfer to real trap images, where clutter, lighting, and camera conditions will differ.

5.1. Assumptions and limitations

The current PoC relies on several assumptions:

- the extracted cutouts are a good enough approximation of moth appearance;
- generated trap-like backgrounds are a reasonable proxy for real trap clutter;
- box-based counting is acceptable for a first internal prototype.

The main limitations are equally important:

- heuristic cutout extraction can fail on dark or cluttered backgrounds
- a synthetic-real domain gap remains
- the detector can overfit to synthetic style, especially because the synthetic dataset is intentionally small
- manual cutout review is useful but labor-intensive and does not scale cleanly
- there is still no real labeled test set for quantitative field evaluation

These limitations are acceptable for a PoC, but they bound what can be claimed from the current results. In particular, more diverse labeled data should improve the detector more than additional architectural complexity at this stage.

5.2. Scaling to a project

The next step is to collect a held-out real dataset with expert labels. Once that exists, the problem can move from a bootstrapping exercise to a reproducible internal computer vision pipeline.

A realistic scale-up plan would be:

- keep the detector as the core model and add a classifier only if false positives remain high;
- use expert review or active learning to focus labeling effort on hard cases;
- include hard negatives such as debris, glue artifacts, and visually similar insects;

- diversify backgrounds, object scales, densities, and pose;
- compare reviewed cutouts against automatic cutouts to quantify the value of manual cleanup;
- compare small and larger YOLO checkpoints before choosing a production baseline;
- introduce explicit dataset/model versioning so training runs remain reproducible.

The critical requirement is a real evaluation set. Without that, the system should be described as a credible proof of concept rather than a validated deployment model.

6. Code availability

The implementation code, including the synthetic data generation scripts, cutout extraction pipeline, training configuration, and inference scripts, is available on GitHub [12].

References

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